NEET UG STANDARD • BIOLOGY TUTORIAL

# Cell: The Unit of Life

**Unit Context:** Cell: Structure and Functions | Comprehensive Concept Guide, Structural Diagrams, Worked Conceptual Examples, and NEET UG Practice Questions

## 1. Learning Objectives & Unit Overview

**After studying this tutorial, you will be able to:**

- Understand the concept of **Reductionist Biology** and the landmark contributions of **G.N. Ramachandran** to structural biology.
- Explain why the cell is the fundamental structural and functional unit of life and trace the formulation of the modern **Cell Theory**.
- Compare cell sizes, shapes, and the fundamental organisation of **prokaryotic** and **eukaryotic** cells.
- Describe the prokaryotic cell envelope, mesosomes, surface appendages (flagella, pili, fimbriae), $70S$ ribosomes, and inclusion bodies.
- Analyse the **Fluid Mosaic Model** of the plasma membrane, membrane transport mechanisms, and the plant cell wall.
- Master the structure and coordinated functions of the **Endomembrane System** (ER, Golgi apparatus, lysosomes, vacuoles), energy-transducing organelles (mitochondria, plastids), ribosomes, cytoskeleton, cilia, flagella, centrosomes, nucleus, and chromosomes.

### Unit Perspective: Cellular Organisation and Reductionist Biology

Biology is the study of living organisms. While describing the form and appearance of organisms highlights their vast diversity, the **cell theory** emphasises the unity underlying this diversity—namely, the cellular organisation of all life forms. A critical mystery of living phenomena (physiological and behavioural processes) is the requirement of the integrity of cellular organisation for living phenomena to be demonstrated or observed.

To study and understand physiological and behavioural processes, scientists take a physico-chemical approach and use cell-free systems to investigate. By analysing living tissues for elements and compounds, we identify what types of organic compounds are present and ask how they carry out gross physiological processes such as digestion, excretion, memory, defense, and recognition. Answering what forms the molecular basis of all physiological processes—and explaining abnormal processes during diseased conditions—by applying the concepts and techniques of physics and chemistry to understand biology is called **Reductionist Biology**.

#### Pioneers in Structural Biology: G.N. Ramachandran (1922 – 2001)

- **Born:** October 8, 1922, in a small town near Cochin on the southwestern coast of India. His father, a professor of mathematics at a local college, shaped his strong interest in mathematics.
- **Education:** Graduated in 1942 as the top-ranking student in the B.Sc. (Honors) Physics course of the University of Madras. Received his Ph.D. from Cambridge University in 1949.
- **Scientific Influence:** While at Cambridge, he met **Linus Pauling** and was deeply influenced by Pauling's publications on models of the α-helix and β-sheet structures, which directed Ramachandran's attention to solving the structure of collagen.
- **Major Contributions:** Founder of the *'Madras school'* of conformational analysis of biopolymers. His discovery of the **triple helical structure of collagen** published in *Nature* in 1954 and his analysis of the

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allowed conformations of proteins through the use of the **'Ramachandran plot'** rank among the most outstanding contributions in structural biology. He passed away at the age of 78 on April 7, 2001.

## 2. What is a Cell? & The Cell Theory

What makes an organism living, or what does a living thing possess that an inanimate object lacks? The answer is the presence of the basic unit of life—the **cell**—in all living organisms. Organisms composed of a single cell are called **unicellular organisms**, while those composed of many cells are called **multicellular organisms**.

Unicellular organisms are capable of:

1. **Independent existence**, and
2. **Performing the essential functions of life**.

Anything less than a complete structure of a cell does not ensure independent living. Hence, the **cell is the fundamental structural and functional unit of all living organisms**.

### Historical Milestones & Cell Theory

- **Antonie Von Leeuwenhoek:** First saw and described a live cell.
- **Robert Brown (1831):** Later discovered and described the nucleus as a cell organelle.
- **Microscopy:** The invention of the microscope and its improvement leading to the electron microscope (in the 1950s) revealed all the structural details of the cell.
- **Matthias Schleiden (1838):** A German botanist who examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant.
- **Theodore Schwann (1839):** A German zoologist who studied different types of animal cells and reported that cells had a thin outer layer which is today known as the *plasma membrane*. Based on his studies on plant tissues, he also concluded that the presence of a **cell wall is a unique character of plant cells**. On this basis, Schwann proposed the hypothesis that the bodies of animals and plants are composed of cells and products of cells.
- **Formulation & Limitation:** Schleiden and Schwann together formulated the cell theory. However, this theory did not explain how new cells were formed.
- **Rudolf Virchow (1855):** First explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

#### Modern Cell Theory

Cell theory as understood today states:

1. All living organisms are composed of cells and products of cells.
2. All cells arise from pre-existing cells (*Omnis cellula-e cellula*).

### An Overview of the Cell

- **Outer Boundaries:** A typical plant cell (such as an onion peel cell) has a distinct **cell wall** as its outer boundary and just within it is the cell membrane. Animal cells (such as human cheek cells) have an outer membrane as the delimiting structure of the cell and lack a cell wall.
- **Nucleus & Genetic Material:** Inside each eukaryotic cell is a dense membrane-bound structure called the **nucleus**, which contains the chromosomes that in turn contain the genetic material, DNA.
  - **Eukaryotic cells:** Cells that have membrane-bound nuclei.
  - **Prokaryotic cells:** Cells that lack a membrane-bound nucleus.

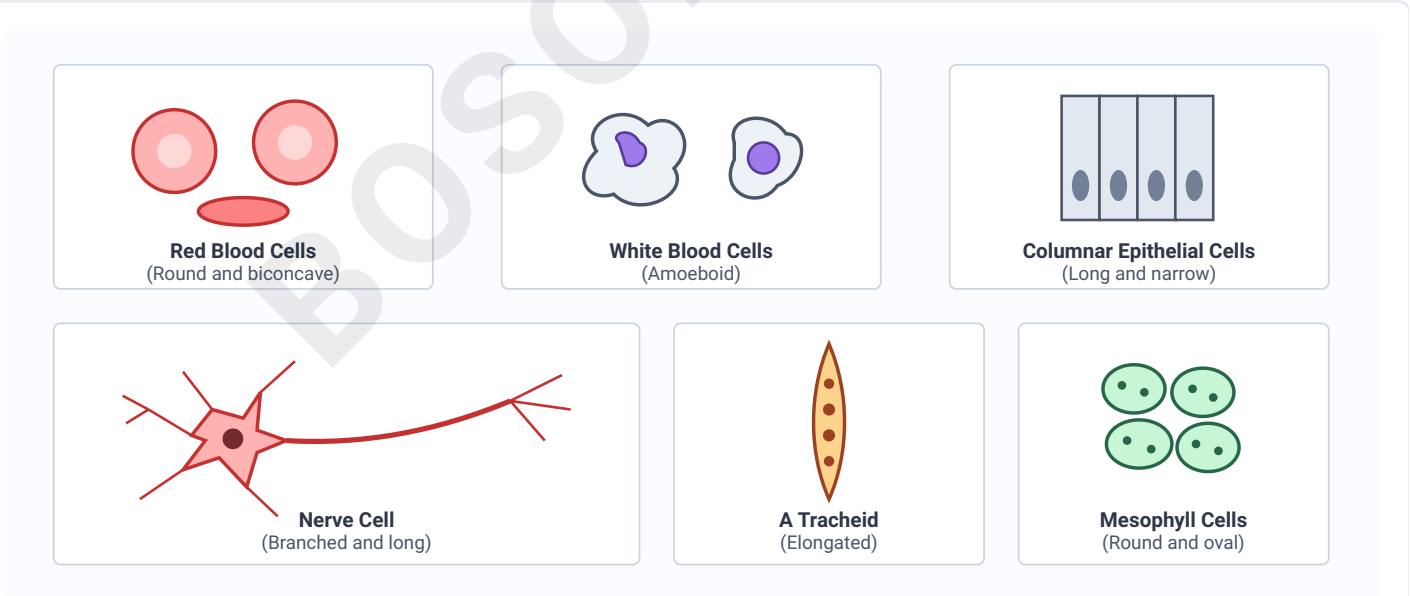
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- **Cytoplasm:** In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. The cytoplasm is the **main arena of cellular activities** in both plant and animal cells; various chemical reactions occur in it to keep the cell in the 'living state'.
- **Membrane-Bound Organelles:** Besides the nucleus, eukaryotic cells have other membrane-bound distinct structures called *organelles* like the endoplasmic reticulum (ER), the Golgi complex, lysosomes, mitochondria, microbodies, and vacuoles. Prokaryotic cells lack such membrane-bound organelles.
- **Non-Membrane Bound Organelles:**
  - **Ribosomes:** Found in *all* cells (both eukaryotic and prokaryotic). Within the cell, ribosomes are found not only in the cytoplasm, but also within two organelles—**chloroplasts** (in plants) and **mitochondria**—and on **rough ER**.
  - **Centrosome:** Present in animal cells; helps in cell division.

### Diversity in Cell Size and Shape

Cells differ greatly in size, shape, and activities. The shape of a cell may vary with the function it performs. Cells may be disc-like, polygonal, columnar, cuboid, thread-like, or even irregular.

| Cell / Organism Type | Characteristic Dimension / Size Feature |
|---|---|
| **Mycoplasmas** (Smallest cells) | Only $0.3\ \mu m$ in length |
| **PPLO** (Pleuro Pneumonia Like Organisms) | About $0.1\ \mu m$ |
| **Viruses** (Non-cellular, for size comparison) | $0.02–0.2\ \mu m$ |
| **Typical Bacteria** (Diagrammatic scale vs general range) | $1–2\ \mu m$ (typical bacterium); general bacterial length $3$ to $5\ \mu m$ |
| **Human Red Blood Cells (RBCs)** | About $7.0\ \mu m$ in diameter (round and biconcave) |
| **Typical Eukaryotic Cell** | $10–20\ \mu m$ |
| **Ostrich Egg** | Largest isolated single cell |
| **Nerve Cells** | Some of the longest cells (branched and long) |




**Tutorial Diagram 1: Diversity of cell shapes correlated with their specialised functions.**

## 3. Prokaryotic Cells

Prokaryotic cells are represented by **bacteria**, **blue-green algae (cyanobacteria)**, **mycoplasma**, and **PPLO (Pleuro Pneumonia Like Organisms)**. They are generally smaller and multiply more rapidly than eukaryotic cells.

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### Basic Shapes of Bacteria

Although prokaryotes exhibit a wide variety of shapes and functions, their cellular organisation is fundamentally similar. The four basic shapes of bacteria are:

1. **Bacillus:** Rod-like
2. **Coccus:** Spherical
3. **Vibrio:** Comma-shaped
4. **Spirillum:** Spiral

### General Organisation of Prokaryotes

- **Cell Wall & Exception:** All prokaryotes have a cell wall surrounding the cell membrane **except in Mycoplasma** (which is cell-wall-less).
- **Cytoplasm & Genetic Material:** The semi-fluid matrix filling the cell is the cytoplasm. There is no well-defined nucleus. The genetic material is basically **naked**, not enveloped by a nuclear membrane.
- **Genomic DNA vs. Plasmid DNA:** In addition to the **genomic DNA** (the single chromosome / circular DNA), many bacteria have small circular DNA outside the genomic DNA called **plasmids**.
  - *Functions of Plasmid DNA:* Confers certain unique phenotypic characters to such bacteria (e.g., **resistance to antibiotics**) and is used to monitor bacterial transformation with foreign DNA.
- **Organelles & Unique Structures:** No membrane-bound organelles (like those in eukaryotes) are found in prokaryotic cells except for **ribosomes** ($70S$). Prokaryotes possess unique reserve deposits called **inclusion bodies** and a specialised differentiated form of cell membrane called the **mesosome** (infoldings of the cell membrane).

### Cell Envelope and its Modifications

Most prokaryotic cells, particularly bacterial cells, have a chemically complex **cell envelope** consisting of a tightly bound three-layered structure that acts together as a **single protective unit** (though each layer performs a distinct function):

1. **Glycocalyx (Outermost layer):** Differs in composition and thickness among different bacteria.
   - **Slime layer:** A loose sheath in some bacteria.
   - **Capsule:** A thick and tough layer in other bacteria.
2. **Cell Wall (Middle layer):** Determines the shape of the cell and provides strong structural support to prevent the bacterium from bursting or collapsing.
3. **Plasma Membrane (Innermost layer):** Selectively permeable in nature and interacts with the outside world. Structurally, this membrane is **similar to that of eukaryotes**.

**Gram Staining Classification:** On the basis of differences in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram, bacteria are classified into two groups:

- **Gram positive bacteria:** Those that take up the Gram stain.
- **Gram negative bacteria:** Those that do not take up the Gram stain.

### Membranous Extensions in Prokaryotes

- **Mesosome:** A special membranous structure formed by the extensions of the plasma membrane into the cell in the form of **vesicles, tubules, and lamellae**.
  - *Functions of Mesosomes:* Help in (1) cell wall formation, (2) DNA replication and distribution to daughter cells, (3) respiration, (4) secretion processes, and (5) increasing the surface area of the plasma membrane and enzymatic content.
- **Chromatophores:** In some prokaryotes like **cyanobacteria**, there are other membranous extensions into the cytoplasm called chromatophores which **contain pigments**.

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### Surface Appendages: Flagella, Pili, and Fimbriae

| Appendage | Structure & Origin | Role / Function |
|---|---|---|
| **Flagella** | Thin filamentous extensions from the **cell wall** in motile bacteria. Composed of three parts: **filament** (longest portion extending from cell surface to outside), **hook**, and **basal body**. | **Motility** (locomotion). Bacteria show a range in number and arrangement of flagella. |
| **Pili** | Elongated tubular structures made of a special protein. | Do **not** play a role in motility. |
| **Fimbriae** | Small bristle-like fibres sprouting out of the cell. | Do **not** play a role in motility. Help attach the bacteria to rocks in streams and also to host tissues. |

### Ribosomes and Inclusion Bodies in Prokaryotes

- **Prokaryotic Ribosomes:** Associated with the **plasma membrane** of the cell. They are about $15\,\text{nm} \times 20\,\text{nm}$ in size and are made of two subunits—$50\text{S}$ and $30\text{S}$ units—which when present together form $70\text{S}$ prokaryotic ribosomes. Ribosomes are the site of protein synthesis.
- **Polyribosomes (Polysome):** Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or a polysome. The ribosomes of a polysome translate the mRNA into proteins.
- **Inclusion Bodies:** Reserve material in prokaryotic cells is stored in the cytoplasm in the form of inclusion bodies. They are **not bound by any membrane system** and lie free in the cytoplasm.
  - *Examples:* Phosphate granules, cyanophycean granules, and glycogen granules.
  - **Gas vacuoles:** Found in blue-green and purple and green photosynthetic bacteria.

## 4. Eukaryotic Cells: Membrane, Wall, and Organelles

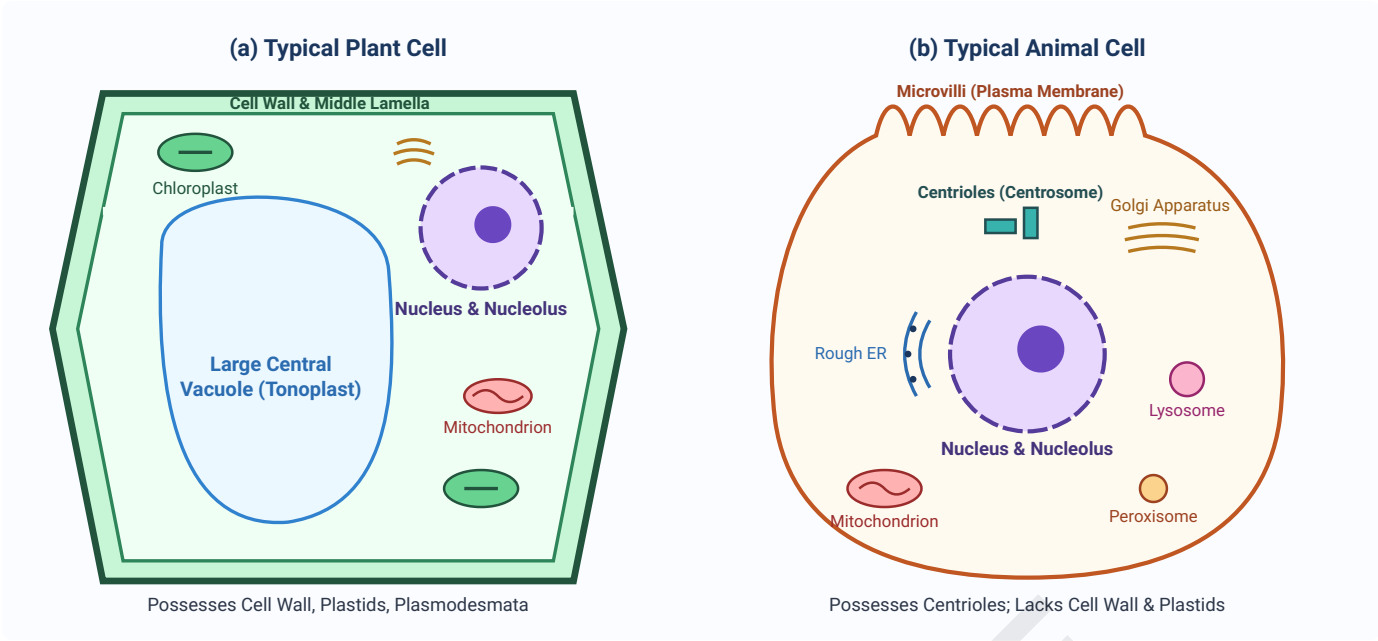
Eukaryotes include all the **protists, plants, animals, and fungi**. Eukaryotic cells feature:

- Extensive compartmentalisation of cytoplasm through the presence of membrane-bound organelles.
- An organised nucleus with a double-layered nuclear envelope.
- A variety of complex locomotory and cytoskeletal structures.
- Genetic material organised into chromosomes.

#### Key Differences Between Plant and Animal Cells

- **Plant cells possess:** Cell wall, plastids, and a large central vacuole (all of which are absent in animal cells).
- **Animal cells possess:** Centrioles (centrosome), which are absent in almost all plant cells.

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**Tutorial Diagram 2: Comparative organisation of (a) a eukaryotic plant cell and (b) a eukaryotic animal cell.**

### 4.1 Cell Membrane (Plasma Membrane)

The detailed structure of the cell membrane was studied only after the advent of the electron microscope in the 1950s. Chemical studies on the cell membrane, especially in **human red blood cells (RBCs)**, enabled scientists to deduce the possible structure of the plasma membrane.

#### Chemical Composition

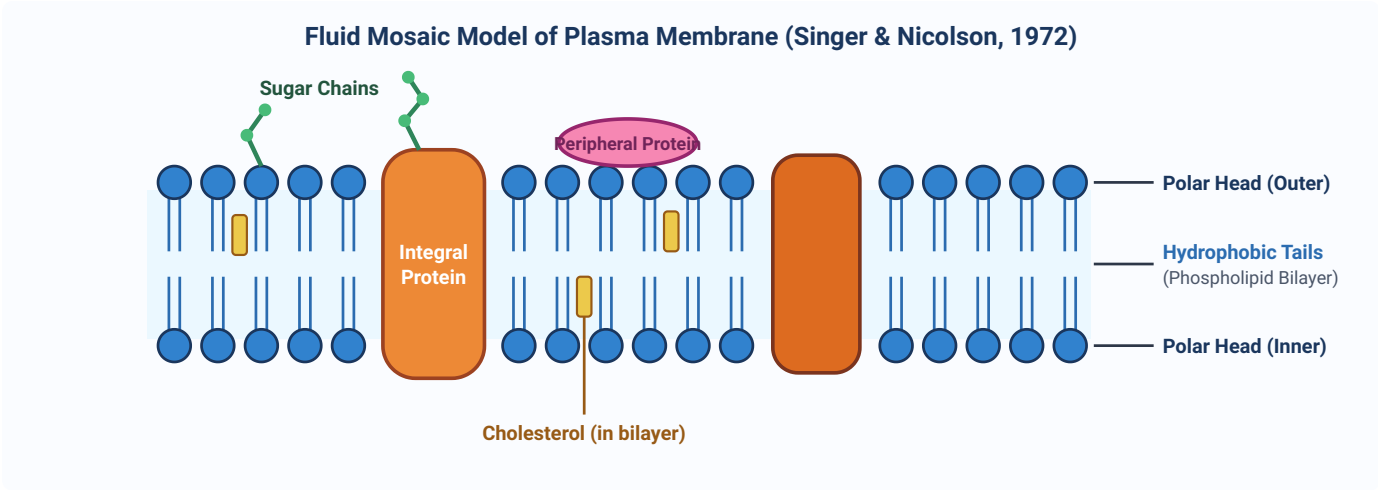
- Composed of **lipids, proteins, and carbohydrates**.
- **Lipids:** Major lipids are **phospholipids** arranged in a **bilayer**.
  - **Polar heads (hydrophilic):** Arranged towards the outer sides.
  - **Hydrophobic tails (nonpolar):** Composed of saturated hydrocarbons arranged towards the inner part. This ensures that the nonpolar tail of saturated hydrocarbons is protected from the aqueous environment.
  - In addition to phospholipids, the membrane also contains **cholesterol**.
- **Protein-to-Lipid Ratio:** Varies considerably in different cell types. In human beings, the membrane of the **erythrocyte (RBC)** has approximately $52\%$ **protein** and $40\%$ **lipids**.
- **Classification of Membrane Proteins (based on ease of extraction):**
  - **Peripheral proteins:** Lie on the surface of the membrane (easily extracted).
  - **Integral proteins:** Partially or totally buried in the membrane.

#### Fluid Mosaic Model (Singer and Nicolson, 1972)

Proposed by **Singer and Nicolson in 1972**, the widely accepted **Fluid Mosaic Model** states that the **quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer**. This ability to move within the membrane is measured as its **fluidity**.

**Functional Importance of Membrane Fluidity:** Essential for cell growth, formation of intercellular junctions, secretion, endocytosis, and cell division.

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**Tutorial Diagram 3: Fluid mosaic model of the plasma membrane showing phospholipid bilayer, cholesterol, integral and peripheral proteins, and carbohydrate (sugar) chains.**

### Transport Across the Plasma Membrane

One of the most important functions of the plasma membrane is the transport of molecules across it. The membrane is **selectively permeable** to some molecules present on either side of it.

| Transport Type | Direction Relative to Gradient | ATP Energy Requirement | Mechanism & Examples |
|---|---|---|---|
| **Passive Transport: Simple Diffusion** | Along the concentration gradient (higher to lower concentration) | No energy required | **Neutral solutes** move across the membrane by simple diffusion. |
| **Passive Transport: Osmosis** | From higher to lower water concentration | No energy required | Movement of **water** by diffusion across the selectively permeable membrane. |
| **Facilitated Diffusion (Carrier-mediated)** | Along the concentration gradient | No energy required | **Polar molecules** cannot pass through the nonpolar lipid bilayer; they require a **carrier protein** of the membrane to facilitate their transport. |
| **Active Transport** | Against the concentration gradient (lower to higher concentration) | Energy-dependent process (**ATP is utilised**) | Pumps ions or molecules against their gradient. Example: $\mathrm{Na^+/K^+}$ **Pump**. |

## 4.2 Cell Wall

A non-living rigid structure called the **cell wall** forms an outer covering for the plasma membrane of **fungi and plants**.

- **Functions of the Cell Wall:**
  1. Gives shape to the cell.
  2. Protects the cell from mechanical damage and infection.
  3. Helps in cell-to-cell interaction.
  4. Provides a barrier to undesirable macromolecules.

- **Chemical Composition by Organism Group:**
  - **Algae:** Cell wall is made of **cellulose, galactans, mannans, and minerals like calcium carbonate (** $\mathrm{CaCO_3}$**)**.
  - **Other Plants:** Consists of **cellulose, hemicellulose, pectins, and proteins**.

- **Layers of the Plant Cell Wall:**
  - **Primary Wall:** The cell wall of a young plant cell; it is **capable of growth**, which gradually diminishes as the cell matures.

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- **Secondary Wall:** Formed as the cell matures on the **inner (towards plasma membrane) side** of the cell.
- **Middle Lamella:** A layer mainly of **calcium pectate** which holds or glues the different neighbouring cells together.

- **Plasmodesmata:** The cell wall and middle lamellae may be traversed by **plasmodesmata**, which connect the cytoplasm of neighbouring cells.

### 4.3 The Endomembrane System

While each membranous organelle is distinct in terms of its structure and function, many of these are considered together as an **endomembrane system** because their **functions are coordinated**.

- **Included in the Endomembrane System (4 organelles):**
  1. Endoplasmic Reticulum (ER)
  2. Golgi complex (Golgi apparatus)
  3. Lysosomes
  4. Vacuoles

- **Excluded from the Endomembrane System: Mitochondria, chloroplasts, and peroxisomes** are *not* considered part of the endomembrane system because their functions are not coordinated with the above components.

#### 4.3.1 The Endoplasmic Reticulum (ER)

Electron microscopic studies of eukaryotic cells reveal a network or reticulum of tiny tubular structures scattered in the cytoplasm called the endoplasmic reticulum (ER). ER divides the intracellular space into two distinct compartments:

- **Luminal compartment:** Space inside the ER.
- **Extra-luminal compartment:** The cytoplasm outside the ER.

| Feature | Rough Endoplasmic Reticulum (RER) | Smooth Endoplasmic Reticulum (SER) |
|---|---|---|
| **Surface Ribosomes** | Bears ribosomes ($80\mathrm{S}$) attached to its outer surface. | Devoid of ribosomes; appears smooth. |
| **Structural Continuity** | Extensive and continuous with the **outer membrane of the nucleus**. | Tubular network continuous with RER. |
| **Primary Function** | Actively involved in **protein synthesis and secretion**. | Major site for **synthesis of lipid**. In animal cells, lipid-like **steroidal hormones** are synthesised in SER. |

#### 4.3.2 Golgi Apparatus

- **Discovery: Camillo Golgi (1898)** first observed densely stained reticular structures near the nucleus, later named Golgi bodies after him.
- **Structure:** Consists of many flat, disc-shaped sacs or **cisternae** of $0.5\ \mu\mathrm{m}$ to $1.0\ \mu\mathrm{m}$ **diameter**, stacked parallel to each other. A varied number of cisternae are present in a Golgi complex.
- **Polarity of Cisternae:** Concentrically arranged near the nucleus with two distinct but interconnected faces:
  - **Convex *cis* face:** The **forming face**.
  - **Concave *trans* face:** The **maturing face**.

- **Coordinated Function with ER:** Principally performs the function of **packaging materials** to be delivered either to intracellular targets or secreted outside the cell.
  - Materials to be packaged in the form of vesicles from the ER fuse with the ***cis* face** of the Golgi apparatus and move towards the maturing face. This explains why the Golgi apparatus remains in close association with the endoplasmic reticulum.
  - A number of proteins synthesised by ribosomes on the endoplasmic reticulum are modified in the cisternae of the Golgi apparatus before they are released from its ***trans* face**.

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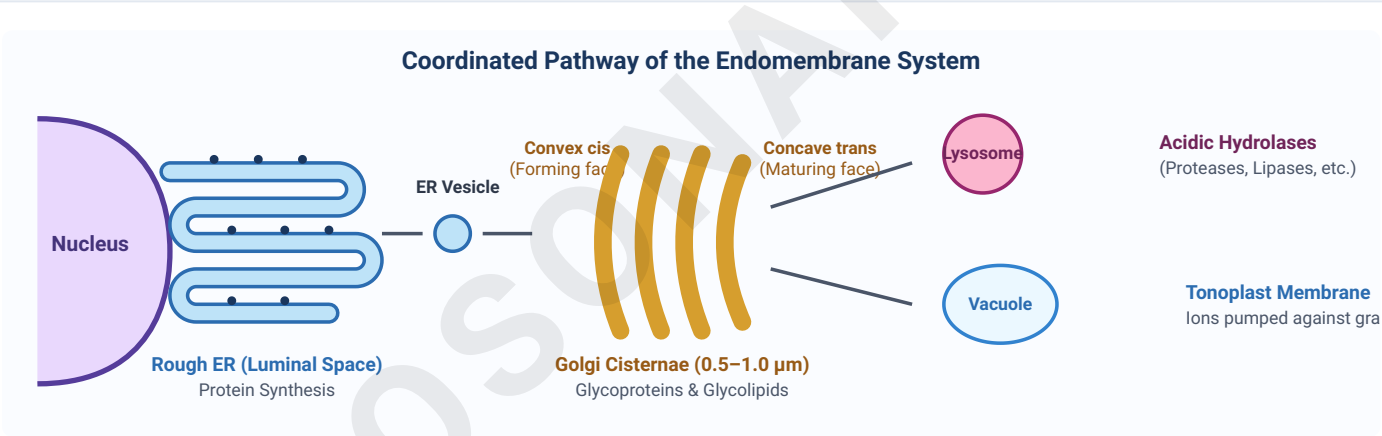
- **Key Biosynthetic Site:** Golgi apparatus is the important site of formation of **glycoproteins and glycolipids**.

### 4.3.3 Lysosomes

- **Origin:** Single membrane-bound vesicular structures formed by the process of **packaging in the Golgi apparatus**.
- **Enzymatic Content:** Isolated lysosomal vesicles are very rich in almost all types of **hydrolytic enzymes** (**hydrolases** – lipases, proteases, carbohydrases) that are **optimally active at the acidic pH**.
- **Function:** Capable of digesting carbohydrates, proteins, lipids, and nucleic acids.

### 4.3.4 Vacuoles

- **Structure & Contents:** Membrane-bound space found in the cytoplasm containing water, sap, excretory product, and other materials not useful for the cell.
- **Tonoplast:** The vacuole is bound by a **single membrane called the tonoplast**. In plant cells, vacuoles can occupy up to $90\%$ **of the volume of the cell**.
- **Active Transport Across Tonoplast:** In plants, the tonoplast facilitates the transport of a number of ions and other materials **against concentration gradients** into the vacuole; hence, their concentration is significantly higher in the vacuole than in the cytoplasm.
- **Specialised Vacuoles:**
  - **Contractile vacuole (in *Amoeba*):** Important for **osmoregulation and excretion**.
  - **Food vacuoles (in many cells, as in protists):** Formed by engulfing food particles.




**Tutorial Diagram 4: Functional coordination among the components of the endomembrane system.**

## 4.4 Mitochondria and Plastids (Semi-Autonomous Double-Membrane Organelles)

### 4.4.1 Mitochondria

- **Visibility & Number:** Unless specifically stained, mitochondria are not easily visible under the microscope. The number of mitochondria per cell is variable depending on the **physiological activity** of the cells.
- **Shape & Dimensions:** Considerably variable in shape and size. Typically **sausage-shaped or cylindrical** having:
  - **Diameter:** $0.2–1.0\,\mu m$ (average $0.5\,\mu m$)
  - **Length:** $1.0–4.1\,\mu m$
- **Double Membrane Organisation:** Outer membrane and inner membrane divide its lumen distinctly into two aqueous compartments: the **outer compartment** and the **inner compartment**.
  - **Outer membrane:** Forms the continuous limiting boundary of the organelle.
  - **Inner compartment (Matrix):** Filled with a dense homogeneous substance called the **matrix**.
  - **Cristae:** The inner membrane forms a number of infoldings called **cristae** (singular: *crista*) towards the matrix. Cristae **increase the surface area**.

- Both membranes have their own specific enzymes associated with mitochondrial function.

- **Function:** Sites of **aerobic respiration** (oxidative phosphorylation). They produce cellular energy in the form of **ATP**, hence they are called the **'power houses' of the cell**.
- **Matrix Components & Division:** The matrix possesses a **single circular DNA molecule**, a few RNA molecules, **ribosomes (**$70\mathrm{S}$**)**, and the components required for the synthesis of proteins. Mitochondria **divide by fission**.

### 4.4.2 Plastids

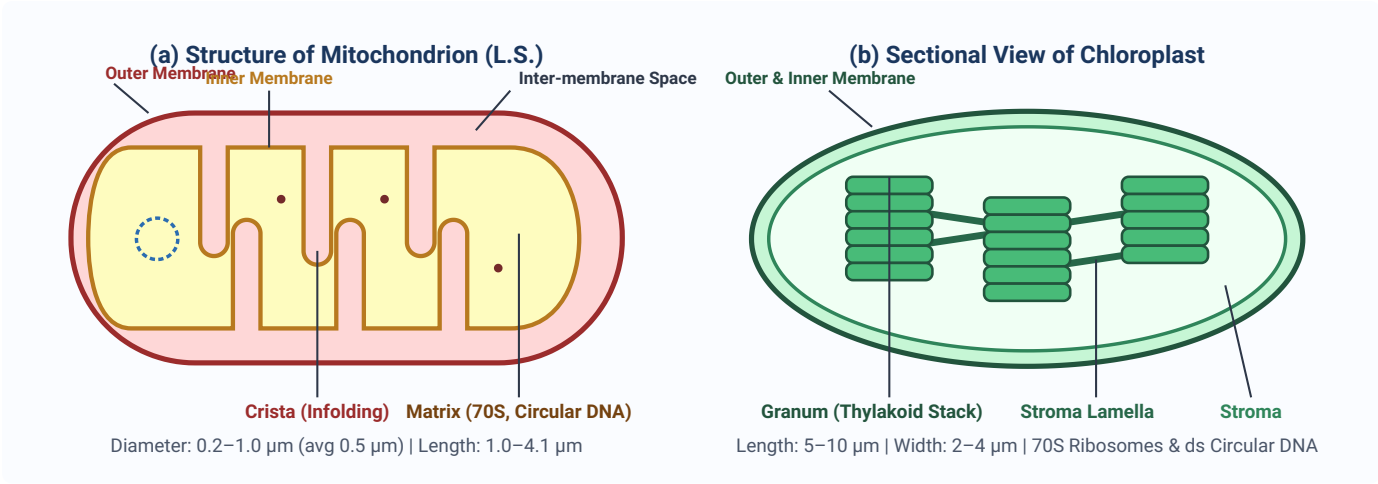
Plastids are found in **all plant cells and in euglenoides**. They are easily observed under the microscope as they are large. Based on the type of pigments they bear, plastids are classified into three types:

1. **Chloroplasts (Green plastids):** Contain **chlorophyll and carotenoid pigments** responsible for trapping light energy essential for photosynthesis.
2. **Chromoplasts (Coloured plastids):** Contain **fat-soluble carotenoid pigments** like **carotene, xanthophylls**, and others. This gives the part of the plant a yellow, orange, or red colour.
3. **Leucoplasts (Colourless plastids):** Colourless plastids of varied shapes and sizes with **stored nutrients**:
   - **Amyloplasts:** Store carbohydrates (starch), e.g., potato.
   - **Elaioplasts:** Store oils and fats.
   - **Aleuroplasts:** Store proteins.

#### Detailed Structure of Chloroplasts

- **Location, Shape & Size:** Majority of the chloroplasts of green plants are found in the **mesophyll cells** of the leaves. They are lens-shaped, oval, spherical, discoid, or even ribbon-like organelles having:
  - **Length:** $5–10\ \mu\mathrm{m}$
  - **Width:** $2–4\ \mu\mathrm{m}$
- **Number per Cell:** Varies from $1$ **per cell** in *Chlamydomonas* (a green alga) to $20–40$ **per cell** in the mesophyll.
- **Membranes & Stroma:** Double membrane-bound. Of the two, the **inner chloroplast membrane is relatively less permeable**. The space limited by the inner membrane of the chloroplast is called the **stroma**.
- **Thylakoids, Grana & Stroma Lamellae:**
  - **Thylakoids:** Organised flattened membranous sacs present in the stroma. The membrane of the thylakoids encloses a space called a **lumen**. **Chlorophyll pigments are present in the thylakoids**.
  - **Grana (singular: granum):** Thylakoids arranged in stacks like the piles of coins (also called *intergranal thylakoids*). Grana are the site of light reactions.
  - **Stroma lamellae:** Flat membranous tubules connecting the thylakoids of the different grana.
- **Stroma Contents:** Contains enzymes required for the **synthesis of carbohydrates and proteins** (site of dark reactions). It also contains **small, double-stranded circular DNA molecules** and $70\mathrm{S}$ **ribosomes** (smaller than the $80\mathrm{S}$ cytoplasmic ribosomes).

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**Tutorial Diagram 5: Double membrane-bound energy-transducing organelles: (a) Mitochondrion and (b) Chloroplast.**

## 5. Ribosomes, Cytoskeleton, Locomotory Structures & Nucleus

### 5.1 Ribosomes

- **Discovery:** Granular structures first observed under the electron microscope as dense particles by **George Palade (1953)**.
- **Composition:** Composed of **ribonucleic acid (RNA) and proteins** and are **not surrounded by any membrane**.
- **Types & Subunits:** Each ribosome has two subunits (larger and smaller):
  - **Eukaryotic ribosomes ($80S$):** Composed of $60S$ (larger) and $40S$ (smaller) subunits.
  - **Prokaryotic ribosomes ($70S$):** Composed of $50S$ (larger) and $30S$ (smaller) subunits. (Also found inside eukaryotic mitochondria and chloroplasts).
- **Meaning of 'S':** Stands for **Svedberg's Unit** (sedimentation coefficient); it is **indirectly a measure of density and size**.

### 5.2 Cytoskeleton

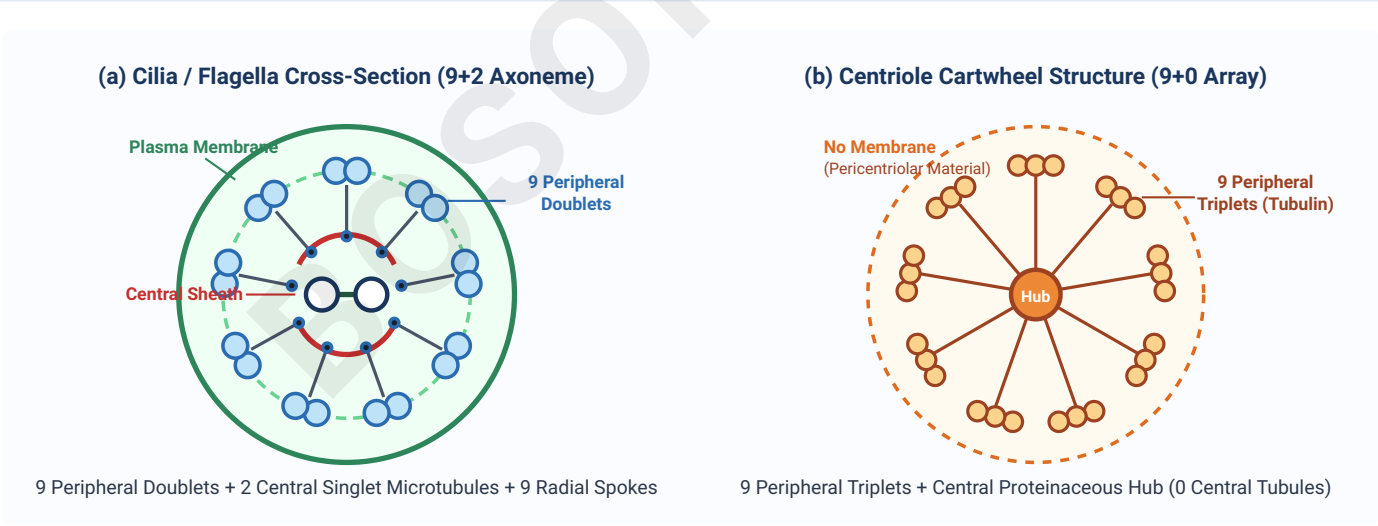
An elaborate network of **filamentous proteinaceous structures** consisting of **microtubules, microfilaments, and intermediate filaments** present in the cytoplasm is collectively referred to as the **cytoskeleton**.

**Functions:** Mechanical support, motility, and maintenance of the shape of the cell.

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## 5.3 Cilia and Flagella vs. Centrosome and Centrioles

| Feature | Cilia and Flagella (Eukaryotic) | Centrosome and Centrioles |
|---|---|---|
| **General Nature & Membrane** | Hair-like outgrowths of the cell membrane; **covered with plasma membrane**. | **Non-membrane bound** organelle containing two cylindrical **centrioles** surrounded by amorphous **pericentriolar materials**. |
| **Orientation / Comparative Action** | **Cilia:** Small structures working like oars, causing movement of either the cell or the surrounding fluid.<br>**Flagella:** Comparatively longer and responsible for cell movement. (Structurally different from prokaryotic flagella). | Both centrioles in a centrosome lie **perpendicular to each other**, in which each has an organisation like a **cartwheel**. |
| **Microtubular Array** | Core is called the **axoneme**, possessing microtubules running parallel to the long axis in a $9+2$ **array**:<br>• $9$ **doublets** of radially arranged peripheral microtubules.<br>• $1$ **pair** $(2)$ of centrally located microtubules. | $9+0$ **arrangement**:<br>• Made up of $9$ **evenly spaced peripheral fibrils of tubulin protein**.<br>• Each peripheral fibril is a **triplet**.<br>• Central microtubule is absent; instead, the central part of the proximal region is proteinaceous and called the **hub**. |
| **Internal Linkages & Spokes** | • Central tubules are connected by **bridges** and enclosed by a **central sheath**.<br>• Central sheath is connected to one of the tubules of each peripheral doublet by a **radial spoke** (total $9$ **radial spokes**).<br>• Peripheral doublets are interconnected by **linkers** (interdoublet bridges). | • Adjacent triplets are linked.<br>• Central proteinaceous **hub** is connected with tubules of the peripheral triplets by **radial spokes** made of protein. |
| **Origin / Functional Role** | Both cilium and flagellum emerge from centriole-like structures called the **basal bodies**. | Centrioles form the **basal body** of cilia or flagella, and **spindle fibres** that give rise to the **spindle apparatus** during cell division in animal cells. |




**Tutorial Diagram 6: Structural comparison between (a) the 9+2 axoneme of eukaryotic cilia/flagella and (b) the 9+0 cartwheel organisation of a centriole.**

## 5.4 Nucleus and Chromosomes

- **Discovery & Naming:**
  - **Robert Brown (1831):** First described the nucleus as a cell organelle.
  - **Flemming:** Gave the name **chromatin** to the material of the nucleus stained by **basic dyes**.
- **Variation in Number of Nuclei per Cell:** Normally, there is only one nucleus per cell, though multinucleated variations occur. Some mature cells even **lack a nucleus (enucleated)**:
  - **Erythrocytes (RBCs) of many mammals**

BOSONANCE

- **Sieve tube cells of vascular plants**

*Conceptual Note:* Mature mammalian erythrocytes and sieve tube cells are considered **living** while functional because they maintain cytoplasmic metabolic activities (and sieve tube elements are closely regulated by adjacent nucleated companion cells).

- **Interphase Nucleus Components:** The nucleus of a cell when it is *not dividing* (interphase nucleus) consists of:
  1. **Nuclear Envelope:** Consists of two parallel membranes with a space between them of $10 \text{ to } 50 \text{ nm}$ called the **perinuclear space**. It forms a barrier between materials inside the nucleus and the cytoplasm. The **outer membrane** usually remains continuous with the endoplasmic reticulum and bears ribosomes on it.
  2. **Nuclear Pores:** At a number of places the nuclear envelope is interrupted by minute pores formed by the **fusion of its two membranes**. Nuclear pores are the passages through which **movement of RNA and protein molecules takes place in both directions** between the nucleus and the cytoplasm.
  3. **Nucleoplasm (Nuclear Matrix):** Contains the nucleolus and chromatin.
  4. **Nucleolus (plural: nucleoli):** One or more spherical structures present in the nucleoplasm. It is **not a membrane-bound structure**; its content is continuous with the rest of the nucleoplasm.
     - *Function:* Site for **active ribosomal RNA (rRNA) synthesis**. Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
  5. **Chromatin:** Loose and indistinct network of nucleoprotein fibres in the interphase nucleus. Chromatin contains **DNA**, some basic proteins called **histones**, some **non-histone proteins**, and also **RNA**.

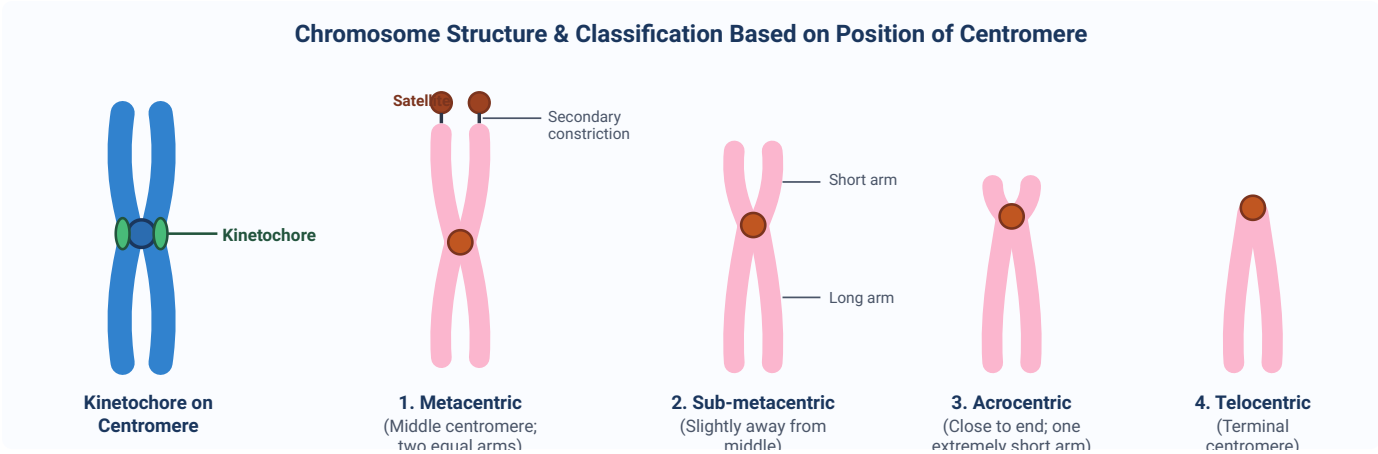
### Chromosomes and Classification Based on Centromere Position

During different stages of cell division, cells show structured **chromosomes** in place of the nucleus. A single human cell has approximately a $2 \text{ m}$ **(two metre) long thread of DNA** distributed among its $46$ **(**$23$ **pairs) chromosomes**.

- **Centromere (Primary Constriction):** Every chromosome (visible only in dividing cells) essentially has a primary constriction or **centromere**, which holds two chromatids of a chromosome together.
- **Kinetochores:** Disc-shaped structures present on the sides of the centromere (where spindle fibres attach during cell division).
- **Secondary Constriction & Satellite:** Sometimes a few chromosomes have **non-staining secondary constrictions at a constant location**. This gives the appearance of a small fragment called the **satellite**.

| Chromosome Type | Position of Centromere | Relative Arm Lengths |
|---|---|---|
| **1. Metacentric** | **Middle** centromere | Forms **two equal arms** of the chromosome. |
| **2. Sub-metacentric** | **Slightly away from the middle** of the chromosome | Results in **one shorter arm and one longer arm**. |
| **3. Acrocentric** | Situated **close to its end** | Forms **one extremely short and one very long arm**. |
| **4. Telocentric** | **Terminal** centromere (at the tip) | Centromere is at the terminal end. |

BOSONANCE




**Tutorial Diagram 7: Chromosome showing kinetochore on centromere, and the four morphological types of chromosomes based on centromere position.**

### 5.5 Microbodies

Many **membrane-bound minute vesicles** called **microbodies** that contain various enzymes are present in **both plant and animal cells**.

## 6. Master Summary & High-Yield Data Tables

For rapid NEET UG revision, review the exact scientists, years, dimensions, and comparative features compiled below.

| Scientist(s) | Year | Landmark Discovery / Contribution |
|---|---|---|
| **Antonie Von Leeuwenhoek** | – | First saw and described a live cell. |
| **Robert Brown** | 1831 | First discovered and described the nucleus as a cell organelle. |
| **Matthias Schleiden** | 1838 | German botanist; observed that all plants are composed of different kinds of cells which form plant tissues. |
| **Theodore Schwann** | 1839 | German zoologist; reported animal cells have a thin outer layer (plasma membrane), concluded presence of cell wall is a unique character of plant cells, and proposed that bodies of animals and plants are composed of cells and products of cells. |
| **Rudolf Virchow** | 1855 | Explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*), giving cell theory its final shape. |
| **Camillo Golgi** | 1898 | First observed densely stained reticular structures near the nucleus (Golgi bodies). |
| **Flemming** | – | Named the nuclear material stained by basic dyes as **chromatin**. |
| **George Palade** | 1953 | First observed ribosomes under the electron microscope as dense granular particles. |
| **G.N. Ramachandran** | 1954 | Founder of the 'Madras school' of conformational analysis of biopolymers; published the triple helical structure of collagen in *Nature* (1954) and developed the Ramachandran plot. |
| **Singer and Nicolson** | 1972 | Proposed the widely accepted **Fluid Mosaic Model** of cell membrane structure. |

BOSONANCE

| Cellular Structure / Feature | Exact Quantitative Value / Dimension |
|---|---|
| Prokaryotic Ribosome size & subunits | $15\,\text{nm} \times 20\,\text{nm}$; 70S (50S + 30S subunits) |
| Eukaryotic Cytoplasmic Ribosome & subunits | 80S (60S + 40S subunits) |
| Human Erythrocyte (RBC) membrane composition | Approximately $52\%$ protein and $40\%$ lipids |
| Golgi Cisternae diameter | $0.5\,\mu\text{m}$ to $1.0\,\mu\text{m}$ |
| Plant Vacuole volume | Can occupy up to $90\%$ of the volume of the plant cell |
| Mitochondrion dimensions | Diameter: $0.2$–$1.0\,\mu\text{m}$ (average $0.5\,\mu\text{m}$); Length: $1.0$–$4.1\,\mu\text{m}$ |
| Chloroplast dimensions & count | Length: $5$–$10\,\mu\text{m}$; Width: $2$–$4\,\mu\text{m}$<br>Count: $1$ per cell in *Chlamydomonas*; $20$–$40$ per cell in leaf mesophyll |
| Perinuclear space width | $10$ to $50\,\text{nm}$ |
| Human cell DNA length & chromosome count | Approximately $2\,\text{m}$ long DNA thread distributed among $46$ ($23$ pairs) chromosomes |

## 7. Worked Conceptual Examples & Guided Discussions

The following worked examples address core conceptual inquiries and analytical comparisons essential for complete mastery of the chapter.

### Worked Example 1: Why is the cell considered the basic unit of life, and how do multicellular organisms exhibit division of labour?

**Discussion:** In unicellular organisms, a single cell is capable of (i) independent existence and (ii) performing all essential functions of life. Anything less than a complete structure of a cell does not ensure independent living. Therefore, the cell is the fundamental structural and functional unit of all living organisms. In multicellular organisms, the body is composed of millions of cells differentiated into various shapes (such as biconcave red blood cells for gas transport, long branched nerve cells for impulse conduction, or oval mesophyll cells rich in chloroplasts for photosynthesis) and tissues, allowing different groups of cells to specialise in distinct physiological functions—a phenomenon known as division of labour.

### Worked Example 2: What is a mesosome in a prokaryotic cell, and what functions does it perform?

**Discussion:** A mesosome is a specialised differentiated form of the cell membrane characteristic of prokaryotes. It is formed by the infoldings/extensions of the plasma membrane into the cell in the form of **vesicles, tubules, and lamellae**. Its functions are:

1. Cell wall formation.
2. DNA replication and distribution to daughter cells.
3. Respiration and secretion processes.
4. Increasing the surface area of the plasma membrane and its enzymatic content.

### Worked Example 3: How do neutral solutes and polar molecules differ in their transport across the plasma membrane?

**Discussion:**

- **Neutral solutes:** Move across the plasma membrane by **simple diffusion** along the concentration gradient (from higher concentration to lower concentration) without any requirement of energy (passive transport).

BOSONANCE

- **Polar molecules:** Cannot pass directly through the nonpolar (hydrophobic) lipid bilayer in the same way. Instead, they require a **carrier protein** of the membrane to facilitate their transport across the membrane (facilitated diffusion). When ions or molecules are transported *against* their concentration gradient (from lower to higher concentration), energy in the form of ATP is utilised via **active transport** (e.g., the $\mathrm{Na}^{+}/\mathrm{K}^{+}$ pump).

**Worked Example 4: Both lysosomes and vacuoles are endomembrane structures, yet they differ in terms of their functions. Explain.**

**Discussion:** Both lysosomes and vacuoles are single membrane-bound organelles whose functions are coordinated with the ER and Golgi apparatus as part of the endomembrane system, yet their physiological roles are distinct:

- **Lysosomes:** Formed by packaging in the Golgi apparatus, they are rich in hydrolytic enzymes (hydrolases—lipases, proteases, carbohydrases) optimally active at acidic pH. Their primary function is the intracellular digestion of carbohydrates, proteins, lipids, and nucleic acids.
- **Vacuoles:** Bound by a single membrane called the **tonoplast**, they store water, sap, excretory products, and other materials not useful for the cell. In plants, the tonoplast actively transports ions and materials against concentration gradients into the vacuole. In *Amoeba*, contractile vacuoles perform osmoregulation and excretion, while in protists, food vacuoles are formed by engulfing food particles.

**Worked Example 5: Compare the two double membrane-bound semi-autonomous cell organelles (mitochondria and chloroplasts).**

**Discussion:**

- **Shared Characteristics:** Both mitochondria and chloroplasts are double membrane-bound organelles excluded from the endomembrane system. Both possess their own circular DNA (single circular DNA in mitochondrial matrix; small double-stranded circular DNA in chloroplast stroma), RNA, $70\mathrm{S}$ ribosomes, and protein synthesis machinery.
- **Distinct Characteristics & Functions:**
  - **Mitochondria:** Present in both plant and animal cells; inner membrane folds inward into **cristae** enclosing the **matrix**; site of **aerobic respiration** and oxidative phosphorylation generating ATP ('power houses' of the cell).
  - **Chloroplasts:** Present in plant cells and euglenoides; inner membrane is relatively less permeable and encloses the **stroma** containing flattened membranous sacs called **thylakoids** stacked into **grana** connected by **stroma lamellae**; contain chlorophyll and carotenoid pigments to trap light energy for **photosynthesis** (light reactions in grana, dark reactions/carbohydrate synthesis in stroma).

## 8. NEET UG Exam-Aligned Practice Questions

*Note: All questions below are original practice questions designed strictly in accordance with the NEET UG examination pattern and syllabus content.*

BOSONANCE

### Practice Question 1 (Single-Choice MCQ)

Which of the following statements regarding the discovery of the cell and the formulation of the Cell Theory is **incorrect**?

(A) Antonie Von Leeuwenhoek first saw and described a live cell, whereas Robert Brown discovered the nucleus.

(B) Matthias Schleiden observed that all plants are composed of different kinds of cells which form the tissues of the plant.

(C) Theodore Schwann concluded that the presence of a cell wall is a unique character of plant cells and also explained how new cells are formed from pre-existing cells.

(D) Rudolf Virchow modified the hypothesis of Schleiden and Schwann by stating *Omnis cellula-e cellula* in $1855$.

**Correct Answer: (C)**

**Worked Solution:** Theodore Schwann ($1839$) reported the thin outer layer (plasma membrane), concluded that the cell wall is a unique character of plant cells, and proposed that the bodies of animals and plants are composed of cells and products of cells. However, the cell theory formulated by Schleiden and Schwann **did not explain how new cells were formed**. It was **Rudolf Virchow ($1855$)** who first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*).

### Practice Question 2 (Single-Choice MCQ)

Regarding the distribution of ribosomes within cells, where are ribosomes found in a typical photosynthetic plant cell?

(A) Only free in the cytoplasm and attached to the Rough Endoplasmic Reticulum.

(B) In the cytoplasm, on the Rough Endoplasmic Reticulum, on the outer nuclear membrane, inside mitochondria, and inside chloroplasts.

(C) Inside the nucleolus, Golgi cisternae, lysosomes, and cytoplasm.

(D) Exclusively associated with the plasma membrane as $70\text{S}$ subunits.

**Correct Answer: (B)**

**Worked Solution:** Ribosomes are non-membrane bound organelles found in all cells. In a eukaryotic plant cell, $80\text{S}$ ribosomes occur freely in the cytoplasm, attached to the surface of the Rough Endoplasmic Reticulum (RER), and on the outer membrane of the nuclear envelope (which is continuous with RER). Additionally, $70\text{S}$ ribosomes are present within two double-membrane organelles: **mitochondria** (in the matrix) and **chloroplasts** (in the stroma).

BOSONANCE

### Practice Question 3 (Multiple-Statement Evaluation)

Read the following statements concerning prokaryotic cells carefully:

1. All prokaryotes have a cell wall surrounding the cell membrane without any exception.
2. Plasmids are small circular DNA molecules outside the genomic DNA that confer unique phenotypic characters such as resistance to antibiotics.
3. Mesosomes are infoldings of the plasma membrane in the form of vesicles, tubules, and lamellae that help in cell wall formation and DNA replication.
4. Pili and fimbriae are surface structures of bacteria that play an active role in bacterial motility alongside flagella.
5. Reserve materials in prokaryotes are stored as membrane-bound inclusion bodies in the cytoplasm.

Which of the above statements are **correct**?

(A) 2 and 3 only

(B) 1, 2, and 3 only

(C) 2, 3, and 4 only

(D) 3 and 5 only

**Correct Answer: (A)**

**Worked Solution:**

- Statement 1 is incorrect: All prokaryotes have a cell wall surrounding the cell membrane **except in Mycoplasma**.
- Statement 2 is correct: Plasmid DNA is small circular extrachromosomal DNA conferring antibiotic resistance and used to monitor bacterial transformation.
- Statement 3 is correct: Mesosomes are extensions of the plasma membrane (vesicles, tubules, lamellae) helping in cell wall formation, DNA replication, respiration, and secretion.
- Statement 4 is incorrect: Pili and fimbriae are surface structures of bacteria, but they **do not play a role in motility**.
- Statement 5 is incorrect: Inclusion bodies are **not bound by any membrane system** and lie free in the cytoplasm.

BOSONANCE

### Practice Question 4 (Matching Type)

Match the structures in **Column I** with their accurate descriptions/functions in **Column II**:

| Column I | Column II |
|---|---|
| (a) Cristae | (i) Flat membranous sacs in the stroma of chloroplasts |
| (b) Cisternae | (ii) Infoldings of the inner membrane in mitochondria |
| (c) Thylakoids | (iii) Disc-shaped sacs ($0.5–1.0\ \mu m$) in Golgi apparatus |
| (d) Chromatophores | (iv) Pigment-containing membranous extensions in cyanobacteria |

Select the correct option:

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(C) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

**Correct Answer: (A)**

**Worked Solution:**

- **Cristae (a)-(ii):** Infoldings of the inner mitochondrial membrane towards the matrix that increase surface area.
- **Cisternae (b)-(iii):** Flat, disc-shaped sacs of $0.5\ \mu m$ to $1.0\ \mu m$ diameter stacked parallel in the Golgi apparatus.
- **Thylakoids (c)-(i):** Organised flattened membranous sacs present in the stroma of chloroplasts.
- **Chromatophores (d)-(iv):** Membranous extensions into the cytoplasm in cyanobacteria that contain pigments.

BOSONANCE

### Practice Question 5 (Assertion–Reason Type)

**Assertion (A):** In human erythrocytes, the plasma membrane contains approximately $52\%$ proteins and $40\%$ lipids, and polar molecules require carrier proteins to cross this membrane.

**Reason (R):** The inner part of the phospholipid bilayer is composed of nonpolar hydrophobic tails of saturated hydrocarbons, which prevents polar molecules from passing through by simple diffusion.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

**Correct Answer: (A)**

**Worked Solution:** Biochemical studies on the human erythrocyte (RBC) membrane showed it has approximately $52\%$ protein and $40\%$ lipids. Because the core of the phospholipid bilayer consists of nonpolar, hydrophobic hydrocarbon tails, polar molecules cannot pass through the nonpolar lipid bilayer on their own and therefore require membrane carrier proteins to facilitate their transport across the membrane. Thus, both (A) and (R) are true, and (R) directly explains why polar molecules need carrier proteins.

### Practice Question 6 (Single-Choice MCQ)

How does the composition of the algal cell wall differ from that of other plants?

(A) Algae have a cell wall made of chitin and pectins, whereas other plants have galactans and mannans.

(B) Algae have a cell wall made of cellulose, galactans, mannans, and minerals like calcium carbonate, while in other plants it consists of cellulose, hemicellulose, pectins, and proteins.

(C) Algae possess a middle lamella made of magnesium carbonate, while other plants have calcium pectate.

(D) Algal cell walls lack cellulose entirely and contain only hemicellulose and calcium pectate.

**Correct Answer: (B)**

**Worked Solution:** Algae have a cell wall made of **cellulose, galactans, mannans, and minerals like calcium carbonate ($\mathrm{CaCO_3}$)**, whereas in other plants it consists of **cellulose, hemicellulose, pectins, and proteins**. In plant tissues, the middle lamella is a layer mainly of calcium pectate that glues neighbouring cells together.

BOSONANCE

### Practice Question 7 (Single-Choice MCQ)

Why are mitochondria, chloroplasts, and peroxisomes **not** considered part of the endomembrane system?

(A) Because all three are surrounded by double membranes.

(B) Because their functions are not coordinated with the components of the endomembrane system (ER, Golgi complex, lysosomes, and vacuoles).

(C) Because they are absent in eukaryotic plant cells.

(D) Because they lack enzymes and cannot synthesise ATP or lipids.

**Correct Answer: (B)**

**Worked Solution:** The endomembrane system includes the endoplasmic reticulum (ER), Golgi complex, lysosomes, and vacuoles because their functions are coordinated. Since the functions of **mitochondria, chloroplasts, and peroxisomes** are **not coordinated** with the above components, they are not considered part of the endomembrane system. (Note: Peroxisomes are single membrane-bound microbodies, not double membrane-bound, making option A false as well).

### Practice Question 8 (Assertion–Reason Type)

**Assertion (A):** The Golgi apparatus remains in close association with the Endoplasmic Reticulum (ER).

**Reason (R):** Materials to be packaged in the form of vesicles from the ER fuse with the convex *cis* (forming) face of the Golgi apparatus, get modified in the cisternae, and are released from the concave *trans* (maturing) face.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

**Correct Answer: (A)**

**Worked Solution:** Materials to be packaged in the form of vesicles from the ER fuse with the *cis* (forming) face of the Golgi apparatus and move towards the maturing (*trans*) face, where proteins synthesised by ribosomes on the ER are modified (e.g., into glycoproteins and glycolipids) before release. This functional continuity directly explains why the Golgi apparatus remains in close association with the ER.

BOSONANCE

### Practice Question 9 (Matching Type)

Match the type of plastid in **Column I** with its specific stored nutrient or pigment in **Column II**:

| Column I | Column II |
|---|---|
| (a) Amyloplasts | (i) Store proteins |
| (b) Elaioplasts | (ii) Fat-soluble carotenoid pigments (carotene, xanthophylls) |
| (c) Aleuroplasts | (iii) Store carbohydrates (starch), e.g., in potato |
| (d) Chromoplasts | (iv) Store oils and fats |

Select the correct option:

(A) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(B) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(D) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

**Correct Answer: (A)**

**Worked Solution:** Leucoplasts are colourless plastids that store nutrients: **Amyloplasts** store carbohydrates/starch (e.g., potato), **Elaioplasts** store oils and fats, and **Aleuroplasts** store proteins. **Chromoplasts** contain fat-soluble carotenoid pigments like carotene and xanthophylls that impart yellow, orange, or red colours to plant parts.

### Practice Question 10 (Single-Choice Analytical MCQ)

What is the total number of microtubules present in the cross-section of a single eukaryotic cilium/flagellum (axoneme) compared to a single centriole?

(A) Axoneme has $11$ microtubules; Centriole has $9$ microtubules.

(B) Axoneme has $20$ microtubules ($9$ doublets $+2$ central); Centriole has $27$ microtubules ($9$ triplets $+0$ central).

(C) Axoneme has $27$ microtubules; Centriole has $20$ microtubules.

(D) Both have $18$ microtubules arranged in a $9+0$ array.

**Correct Answer: (B)**

**Worked Solution:**

- In the **axoneme of a eukaryotic cilium or flagellum ($9+2$ array)**, there are $9$ peripheral doublets ($9 \times 2 = 18$ microtubules) and $1$ pair of centrally located microtubules ($2$ microtubules), giving a total of $18+2=20$ microtubules (along with $9$ radial spokes).
- In a **centriole ($9+0$ cartwheel organisation)**, there are $9$ evenly spaced peripheral triplets of tubulin ($9 \times 3 = 27$ microtubules) and no central microtubules (instead, there is a proteinaceous central hub connected by $9$ radial spokes).

BOSONANCE

### Practice Question 11 (Multiple-Statement Evaluation)

Consider the following statements regarding the eukaryotic nucleus:

1. The nuclear envelope consists of two parallel membranes separated by a perinuclear space of $10 \text{ to } 50 \, \text{nm}$.
2. Nuclear pores are formed by the fusion of only the inner nuclear membrane with the nucleolus, allowing unidirectional movement of DNA into the cytoplasm.
3. The nucleolus is a non-membrane bound spherical structure in the nucleoplasm that serves as the site for active ribosomal RNA (rRNA) synthesis.
4. Mature erythrocytes of many mammals and sieve tube cells of vascular plants lack a nucleus.

Which of the above statements are **correct**?

(A) 1, 2, and 3 only

(B) 1, 3, and 4 only

(C) 2 and 4 only

(D) 1, 2, 3, and 4

**Correct Answer: (B)**

**Worked Solution:** Statements 1, 3, and 4 are completely accurate. Statement 2 is incorrect because **nuclear pores** are formed by the **fusion of the two membranes of the nuclear envelope**, and they act as passages through which **movement of RNA and protein molecules takes place in both directions** between the nucleus and the cytoplasm.

### Practice Question 12 (Single-Choice MCQ)

Identify the correct pairing of chromosome type, centromere position, and structural features:

(A) **Sub-metacentric:** Centromere close to the end, forming one extremely short and one very long arm.

(B) **Acrocentric:** Centromere slightly away from the middle, resulting in one shorter arm and one longer arm.

(C) **Satellite (SAT) region:** Appears due to a non-staining secondary constriction at a constant location on a few chromosomes.

(D) **Kinetochore:** Terminal region of a telocentric chromosome that synthesises histone proteins.

**Correct Answer: (C)**

**Worked Solution:** Sometimes a few chromosomes have **non-staining secondary constrictions at a constant location**, which gives the appearance of a small fragment called the **satellite**. (Option A describes an acrocentric chromosome; Option B describes a sub-metacentric chromosome; Option D is incorrect because kinetochores are disc-shaped structures present on the sides of the primary constriction/centromere).

BOSONANCE

## 9. Common Exam Pitfalls & Final Revision Checklist

### Common NEET UG Pitfalls to Avoid

- **Pitfall 1 — Robert Brown vs. Leeuwenhoek:** Do *not* mark Robert Brown as the discoverer of the cell. Antonie Von Leeuwenhoek first saw and described a live cell; Robert Brown $(1831)$ discovered the **nucleus**.
- **Pitfall 2 — Cell Wall Exception in Prokaryotes:** All prokaryotes have a cell wall surrounding the cell membrane **except Mycoplasma**.
- **Pitfall 3 — Pili and Fimbriae vs. Flagella:** Pili and fimbriae are surface structures in bacteria, but they **do NOT play any role in motility**. Fimbriae help in attachment to rocks in streams and host tissues.
- **Pitfall 4 — Inclusion Bodies & Nucleolus Membrane Status:** Neither prokaryotic **inclusion bodies** nor the eukaryotic **nucleolus** nor **ribosomes** nor **centrosomes** are bound by a membrane.
- **Pitfall 5 — Tonoplast Ion Concentration:** In plants, the tonoplast transports ions **against concentration gradients** into the vacuole; therefore, ion concentration is **significantly higher in the vacuole than in the cytoplasm**.
- **Pitfall 6 — Svedberg's Unit ('S') is Not Additive:** $50\text{S} + 30\text{S} = 70\text{S}$ (prokaryotic/organellar) and $60\text{S} + 40\text{S} = 80\text{S}$ (eukaryotic cytoplasmic) because 'S' measures the sedimentation coefficient (an indirect measure of density and size), not simple mass addition.
- **Pitfall 7 — Centromere vs. Kinetochore:** The **centromere** is the primary constriction holding two chromatids together, whereas **kinetochores** are the disc-shaped structures on the sides of the centromere.

### Concise Final Recap

- **Cell Theory:** Formulated by Schleiden $(1838)$ and Schwann $(1839)$; completed by Rudolf Virchow ( $1855$, *Omnis cellula-e cellula*).
- **Prokaryotes:** Naked circular genomic DNA + plasmids; $70\text{S}$ ribosomes at plasma membrane; three-layered cell envelope (glycocalyx $\rightarrow$ cell wall $\rightarrow$ plasma membrane); mesosomes (vesicles, tubules, lamellae); chromatophores in cyanobacteria; inclusion bodies free in cytoplasm.
- **Plasma Membrane:** Fluid Mosaic Model (Singer & Nicolson, $1972$); phospholipid bilayer + cholesterol + integral/peripheral proteins + carbohydrates ($52\%$ protein, $40\%$ lipid in human RBC).
- **Endomembrane System:** ER (RER for protein synthesis/secretion; SER for lipids/steroidal hormones) $\rightarrow$ Golgi apparatus (*cis* forming to *trans* maturing face; glycoproteins & glycolipids) $\rightarrow$ Lysosomes (acidic hydrolases) & Vacuoles (tonoplast). Excludes mitochondria, chloroplasts, and peroxisomes.
- **Mitochondria & Plastids:** Double membrane-bound, contain circular DNA and $70\text{S}$ ribosomes. Mitochondria have cristae and perform aerobic respiration (ATP generation). Chloroplasts have grana (light reactions) and stroma (dark reactions, carbohydrate/protein synthesis).
- **Cytoskeleton & Motility:** Microtubules, microfilaments, intermediate filaments. Cilia/flagella have a membrane-covered $9 + 2$ axoneme emerging from basal bodies; centrioles have a non-membranous $9 + 0$ cartwheel of tubulin triplets.
- **Nucleus & Chromosomes:** Double membrane with perinuclear space $(10\text{–}50\ \text{nm})$ and bidirectional nuclear pores; nucleoplasm contains chromatin (DNA + histones + non-histones + RNA) and nucleolus (rRNA synthesis). Chromosomes are classified by centromere position into metacentric, sub-metacentric, acrocentric, and telocentric.

BOSONANCE